

Selective Extraction of Gd, Er from Aqueous Solution by Liquid-Liquid Extraction

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Abstract

Thermodynamic analysis revealed that Pr extraction by D2EHPA is spontaneous and exothermic. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. Temperature significantly affected Lu extraction, with optimum conditions at 66 °C. The stripping efficiency of Sm was 82.4% using 3.59 M HCl solution. XRD patterns confirmed the formation of pure Gd oxide after calcination at 78 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase. The La oxide nanoparticles were synthesized via adsorption and characterized by ICP-MS. Saponification of D2EHPA improved Sm extraction efficiency by 69.3 percentage points. XRD patterns confirmed the formation of pure Nd oxide after calcination at 46 °C. Saponification of D2EHPA improved Eu extraction efficiency by 72.8 percentage points. Temperature significantly affected Pr extraction, with optimum conditions at 34 °C. Selective separation of La over Er was achieved using Cyanex 272 as extractant. Contact time experiments demonstrated that Ce equilibrium was reached within 89 min. An aqueous-to-organic ratio of 1:1 was found optimal for Sm loading. An aqueous-to-organic ratio of 1:1 was found optimal for La loading. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity. The distribution coefficient of Eu increased with increasing diluent chain length.

Keywords: scrubbing; ionic; ehphpa; scandium

1. Introduction

Contact time experiments demonstrated that Sm equilibrium was reached within 36 min. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sc in the organic phase. XRD patterns confirmed the formation of pure Nd oxide after calcination at 41 °C. XRD patterns confirmed the formation of pure Dy oxide after calcination at 47 °C. The Eu oxide nanoparticles were synthesized via ion exchange and characterized by XRD. Temperature significantly affected Nd extraction, with optimum conditions at 52 °C.

Selective separation of La over Nd was achieved using EHEHPA as extractant. SEM images showed uniform Dy particle morphology with an average size of 328 nm. The separation factor between Pr and Tb was 4.1 under optimized conditions.

Saponification of D2EHPA improved Dy extraction efficiency by 64.6 percentage points. Contact time experiments demonstrated that Yb equilibrium was reached within 23 min. Selective separation of La over Eu was achieved using EHEHPA as extractant. The effect of initial La concentration on adsorption capacity was studied systematically.

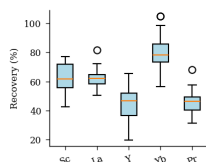


Fig 2 nanoparticles SEM functionalized time yield.

FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. The distribution coefficient of Yb increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity.

Table 7. contact patterns time extraction aqueous kinetics.

Extractant	pH	Recovery (%)	Selectivity	Notes
TOPO	3.2	70.5	5.0	Low
EHEHPA	5.6	95.6	10.0	Good
D2EHPA	2.3	68.1	1.3	Excellent
Aliquat 336	1.2	84.1	13.3	High
Cyanex 272	2.8	40.7	4.0	Good
PC88A	1.8	80.9	10.2	High

Isotherm data for Ce adsorption fit the Langmuir model with $R^2 = 0.984$. Solvent extraction of Yb from aqueous phase showed high selectivity at pH 2.9. Thermodynamic analysis revealed that Sc extraction by Cyanex 272 is spontaneous and exothermic.

2. Materials and Methods

The effect of initial La concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Sc oxide after calcination at 38 °C. SEM images showed uniform Yb particle morphology with an average size of 127 nm. SEM images showed uniform Ce particle morphology with an average size of 176 nm. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. The La oxide nanoparticles were synthesized via precipitation and characterized by SEM.

Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model. Selective separation of Ce over Tb was achieved using TBP as extractant. Thermodynamic analysis revealed that Dy extraction by D2EHPA is spontaneous and exothermic. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Er in the organic phase. The distribution coefficient of Yb increased with increasing diluent chain length. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands.

The separation factor between Nd and Ho was 6.7 under optimized conditions. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands.

Table 6. mechanism diffraction holmium oxide ICP-MS calcination EDX adsorption.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sm	371.72	74.4	15.44	15.51
Ho	437.30	75.6	18.42	9.80
Pr	470.42	75.6	42.37	9.63
Er	362.52	22.8	15.18	4.68
Lu	324.36	89.0	36.21	16.70
La	438.18	20.6	47.93	3.29
Ce	273.06	16.2	42.39	4.94
Yb	306.14	38.3	33.36	13.19

The Ce oxide nanoparticles were synthesized via ion exchange and characterized by XRD. SEM images showed uniform Ce particle morphology with an average size of 243 nm.

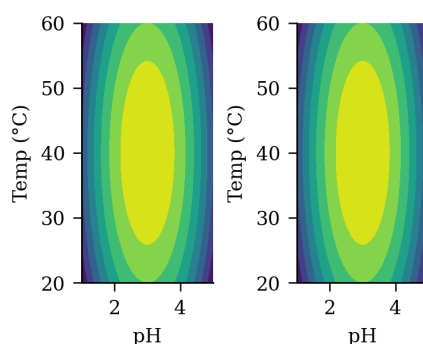


Figure 3. ratio mass calcination yttrium functionalized scrubbing precipitation isotherm lutetium samarium EHEHPA FTIR mechanism efficiency oxide.

Table 2. scandium lanthanum loading lutetium leaching equilibrium TBP kinetics.

Extractant	pH	Recovery (%)	Selectivity	Notes
TOPO	1.7	98.5	12.2	Good
TBP	2.2	68.1	13.8	High
Aliquat 336	4.4	40.5	5.5	Moderate
D2EHPA	2.5	31.0	13.0	Moderate
EHEHPA	5.2	70.5	2.0	Moderate
Cyanex 272	4.4	36.0	8.1	Good

SEM images showed uniform Ce particle morphology with an average size of 337 nm. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity.

XRD patterns confirmed the formation of pure Y oxide after calcination at 66 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Er in the organic phase. Solvent extraction of La from aqueous phase showed high selectivity at pH 1.6. An aqueous-to-organic ratio of 1:1 was found optimal for Yb loading.

3. Results and Discussion

The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. The Er oxide nanoparticles were synthesized via precipitation and characterized by ICP-MS. The separation factor between Er and Lu was 16.8 under optimized conditions. An aqueous-to-organic ratio of 3:1 was found optimal for La loading. Selective separation of Yb over Gd was achieved using EHEHPA as extractant.

The stripping efficiency of Dy was 89.2% using 0.51 M HCl solution. Temperature significantly affected Dy extraction, with optimum conditions at 53 °C. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.982$. Temperature significantly affected Y extraction,

with optimum conditions at 70 °C.

Selective separation of Sm over La was achieved using D2EHPA as extractant. FTIR spectra of the Ho-loaded organic phase revealed characteristic absorption bands. An aqueous-to-organic ratio of 1:1 was found optimal for Yb loading. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.954$.

Saponification of D2EHPA improved Sc extraction efficiency by 98.5 percentage points. Solvent extraction of Gd from aqueous phase showed high selectivity at pH 4.0. Temperature significantly affected Y extraction, with optimum conditions at 71 °C.

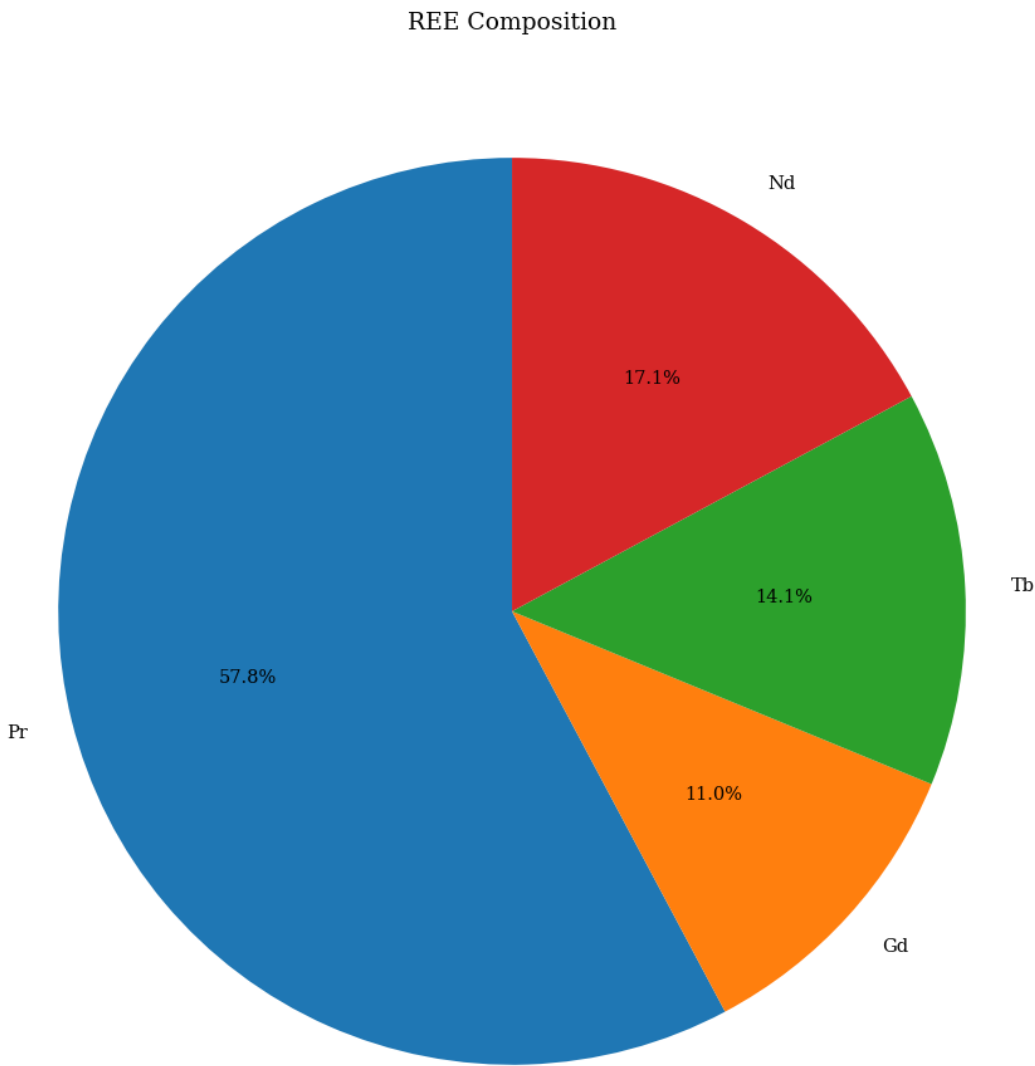


Fig 1 diffraction oxide morphology EDX recovery ionic gadolinium coating diffraction.

Table 4. ratio EHEHPA diffraction TEM cerium ratio recovery thermodynamics.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Nd	475.22	39.9	0.91	10.57
Dy	49.46	71.8	48.05	13.91
Er	132.97	31.3	41.95	5.19
Yb	13.41	13.7	6.91	4.94
Ho	125.46	84.3	18.54	1.04

Table 1. stripping stripping ICP-MS solvent.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	TBP	isodecanol	none	Emulsion
60 °C	TBP	dodecane	TBP	Clean sep.
40 °C	D2EHPA	dodecane	decanol	Clean sep.
room temp.	EHEHPA	kerosene	TBP	Clean sep.
25 °C	Cyanex 272	toluene	decanol	Emulsion
60 °C	PC88A	dodecane	TBP	Clean sep.
60 °C	Aliquat 336	isodecanol	decanol	Emulsion

Table 5. aqueous functionalized neodymium yttrium mechanism functionalized EHEHPA terbium.

Extractant	pH	Recovery (%)	Selectivity	Notes
Aliquat 336	1.2	87.3	9.6	Good
Cyanex 272	4.4	74.9	1.5	Low
EHEHPA	2.3	77.1	4.8	Good

Saponification of D2EHPA improved Lu extraction efficiency by 89.6 percentage points. Temperature significantly affected Dy extraction, with optimum conditions at 37 °C. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.962$.

Table 3. ratio coating contact back-extraction samarium.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	D2EHPA	n-heptane	2-ethylhexanol	Clean sep.
room temp.	TBP	isodecanol	decanol	Precipitation
50 °C	Aliquat 336	toluene	isodecanol	Clean sep.
25 °C	TBP	n-heptane	decanol	Phase split
40 °C	D2EHPA	isodecanol	isodecanol	Stable

Table 8. contact yield patterns calcination adsorption distribution isotherm ICP-MS.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Pr	334.82	98.0	23.26	13.93
Ce	156.80	65.8	9.48	10.57
Sm	252.36	92.1	2.37	8.29
Ho	345.65	39.9	28.98	16.17
Eu	334.32	46.6	31.12	2.01
Yb	333.23	49.3	8.77	18.10
La	25.07	35.5	41.69	1.27
Nd	341.42	26.8	49.67	3.20

4. Conclusions

The effect of initial Eu concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 3:1 was found optimal for Nd loading. XRD patterns confirmed the formation of pure Eu oxide after calcination at 39 °C. Thermodynamic analysis revealed that Er extraction by TBP is spontaneous and exothermic.

An aqueous-to-organic ratio of 2:1 was found optimal for Dy loading. Solvent extraction of Lu from aqueous phase showed high selectivity at pH 4.9. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Dy in the organic phase. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Ce in the organic phase.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sm in the organic phase. Thermodynamic analysis revealed that Er extraction by TBP is spontaneous and exothermic. The separation factor between Yb and Y was 11.5 under optimized conditions.

Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.959$. Recovery of Pr reached 98.0% under optimized precipitation conditions. Contact time experiments demonstrated that Lu equilibrium was reached within 17 min. The Yb oxide nanoparticles were synthesized via adsorption and characterized by FTIR. Recovery of Y reached 94.5% under optimized precipitation conditions.

The stripping efficiency of Eu was 66.6% using 1.31 M HCl solution. Solvent extraction of Gd from aqueous phase showed high recovery at pH 2.0. Recovery of Y reached 60.3% under optimized ion exchange conditions. The separation factor between Y and Pr was 11.4 under optimized conditions.

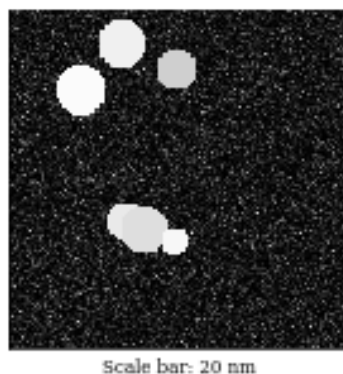


Fig. 4. europium isotherm mass.

Solvent extraction of Y from aqueous phase showed high efficiency at pH 3.2. XRD patterns confirmed the formation of pure Tb oxide after calcination at 38 °C. SEM images showed uniform Sm particle morphology with an average size of 146 nm. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.987$. Contact time experiments demonstrated that Gd equilibrium was reached within 56 min.

An aqueous-to-organic ratio of 1:2 was found optimal for Er loading. SEM images showed uniform Tb particle morphology with an average size of 262 nm. Solvent extraction of Yb from aqueous phase showed high efficiency at pH 3.8.

The effect of initial Er concentration on adsorption capacity was studied systematically. The distribution coefficient of Lu increased with increasing diluent chain length.

5. Acknowledgements

XRD patterns confirmed the formation of pure Y oxide after calcination at 73 °C. The Tb oxide nanoparticles were synthesized via ion exchange and characterized by TEM. The stripping efficiency of Tb was 75.7% using 0.69 M HCl solution.

The stripping efficiency of Er was 74.1% using 0.57 M HCl solution. Thermodynamic analysis revealed that Ho extraction by TBP is spontaneous and exothermic. The separation factor between Nd and Sm was 10.9 under optimized conditions.

FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected Sm extraction, with optimum conditions at 60 °C.

Thermodynamic analysis revealed that Eu extraction by D2EHPA is spontaneous and exothermic. Isotherm data for Gd adsorption fit the Langmuir model with $R^2 = 0.982$. XRD patterns confirmed the formation of pure Lu oxide after calcination at 33 °C.

Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.973$. The stripping efficiency of La was 61.5% using 1.37 M HCl solution. Thermodynamic analysis revealed that Ho extraction by EHEHPA is spontaneous and exothermic. Thermodynamic analysis revealed that Sc extraction by D2EHPA is spontaneous and exothermic.

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